Shahid Beheshti University
Faculty of Mathematical Sciences

Clustring and Vanila Neural Networks

Sepehr Barekati
404422029

Course: Data Mining — *Professor:* Dr. Farahani

---

1 Introduction

This report details the implementation of a machine learning pipeline on the Movies Metadata dataset. The analysis is divided into three main phases: Exploratory Data Analysis (EDA) and preprocessing, Clustering Analysis using K-Means and Hierarchical algorithms, and Predictive Modeling utilizing a deep fully connected neural network to estimate movie revenue.

2 Exploratory Data Analysis & Preprocessing

2.1 Data Inspection and Cleaning

The dataset underwent initial inspection to generate summary statistics. Rows containing missing values (NaNs) were strictly removed to maintain data integrity for downstream algorithms.

2.2 Visualization and Findings

Visualizations were generated to understand feature distributions and relationships. The findings from these plots are summarized in Table 1.

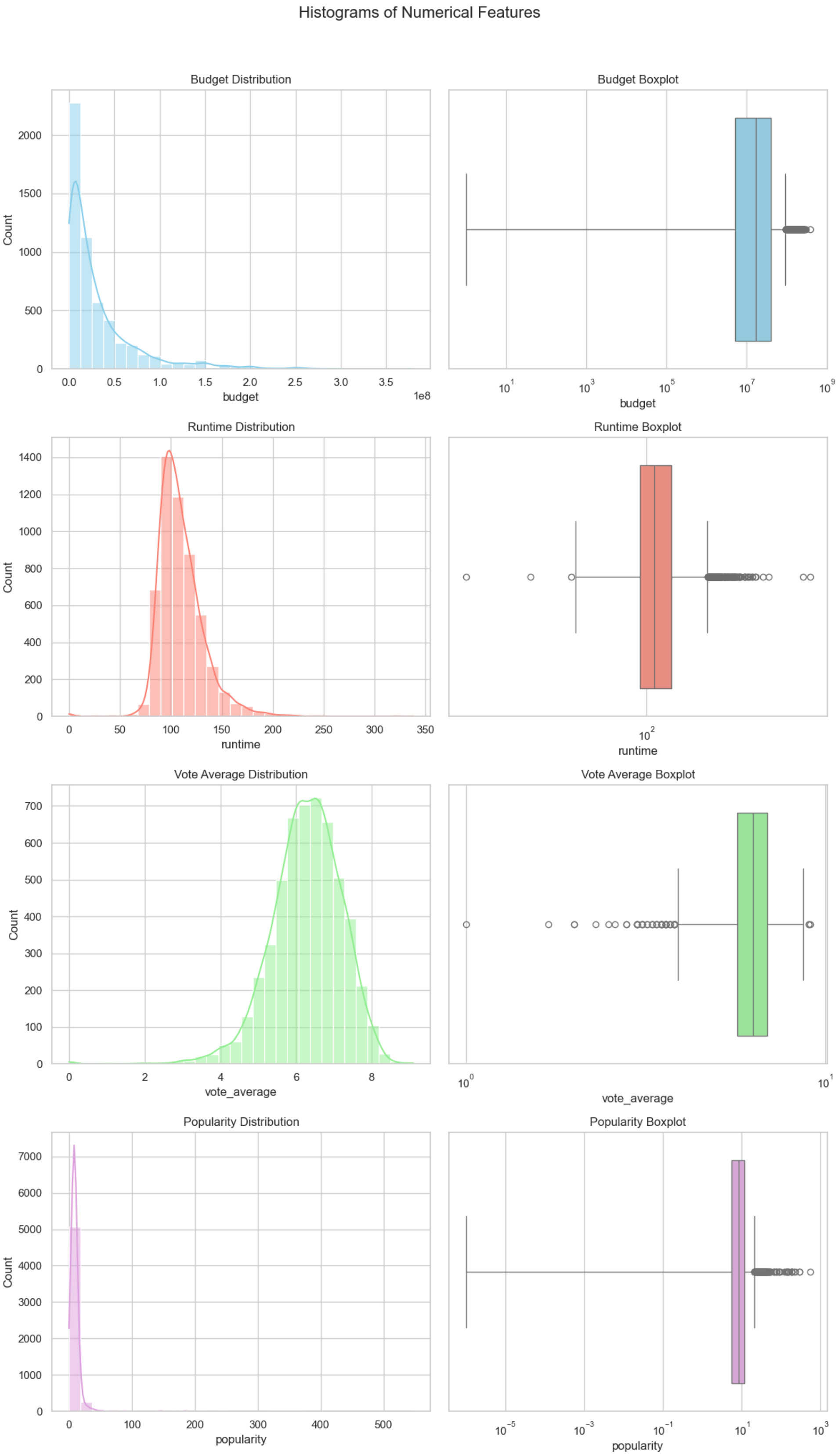

Figure 1: Histograms and Boxplots of Budget, Runtime, Vote_average, and Popularity

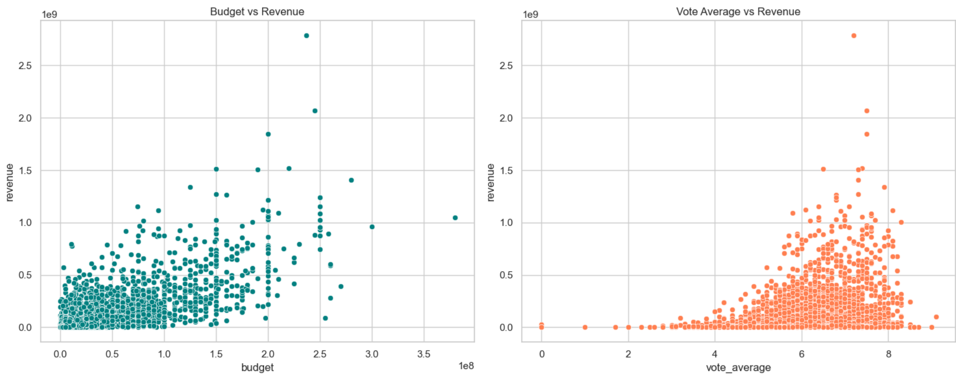

Figure 2: Scatter Plots: Budget and Vote_average Relationships with Revenue

Table 1: Summary of EDA Visualizations

Plot Type	**Features Evaluated**	**Key Findings**
Histograms	Budget, Runtime, Popularity	Financial metrics and popularity are heavily right-skewed, indicating blockbuster dominance.
Histograms	Vote Average	Exhibits a near-normal, bell-curve distribution.
Boxplots	All Numerical Features	Severe outliers detected on the high end of Budget, Popularity, and Revenue.
Scatter Plots	Budget vs. Revenue	Positive correlation observed; higher budgets generally yield higher revenues.
Heatmap	Correlation Matrix	Highlights linear relationships between specific input features and the target variable.

3 Clustering Analysis

The objective of this phase was to group movies based on three normalized features: `budget`, `runtime`, and `vote_average`. The target variable, `revenue`, was excluded to prevent data leakage. Feature scaling via `StandardScaler` was applied prior to distance calculations.

3.1 K-Means Clustering

K-Means was evaluated for $k = 2$ to 10. The elbow method (using Inertia) and Silhouette Scores were utilized to determine the optimal grouping.

3.2 Hierarchical Clustering

Agglomerative clustering (Ward linkage, Euclidean metric) was similarly evaluated. Since hierarchical clustering does not natively produce inertia, a supplementary K-Means model was fitted per k strictly to compute the within-cluster sum of squares.

3.3 Optimal Clusters Selection

The optimal number of clusters for each algorithm was determined mathematically via the plotted metrics. The results are summarized in Table 2.

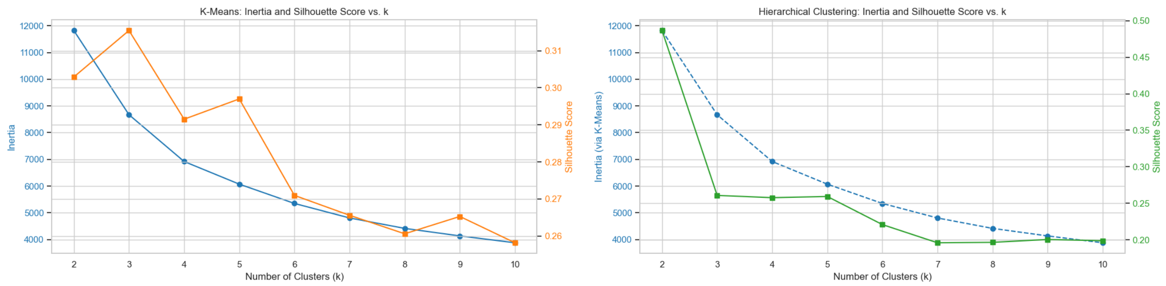

Figure 3: Inertia and Silhouette Scores vs. k for K-Means (Left) and Hierarchical (Right)

Table 2: Clustering Optimization Results

Algorithm	**Optimal k**	**Primary Indicator**	**Silhouette Score**
K-Means	3	Global peak on Silhouette, distinct elbow	0.318
Hierarchical	2	Massive global peak on Silhouette	0.490

4 Neural Network for Revenue Prediction

4.1 Data Splitting and Scaling

The dataset was divided into Training (70%), Validation (15%), and Test (15%) subsets. To stabilize the gradients during backpropagation and prevent the model from failing (yielding a negative R^2), both the input features and the massive target variable (`revenue`) were standardized using independent `StandardScaler` instances fit strictly on the training data.

4.2 Architecture and Configuration

A sequential deep fully connected neural network was constructed. The detailed architecture and training configuration are outlined in Table 3.

Table 3: Neural Network Architecture and Training Configuration

Component	Specification
Input Layer	Neurons equal to feature count
Hidden Layer 1	64 Neurons, ReLU Activation
Hidden Layer 2	32 Neurons, ReLU Activation
Output Layer	1 Neuron, Linear Activation
Total Trainable Params	2,625
Loss Function	Mean Squared Error (MSE)
Optimizer	Adam
Epochs	100
Batch Size	32

4.3 Final Evaluation

Following training, the model's predictions on the unseen test set were inverse-transformed back to original dollar amounts to calculate the final metrics.

Table 4: Final Model Evaluation Metrics (Test Set)

Metric	Value
Mean Squared Error (MSE)	$5,727,811,650,471,147.00$
Root Mean Squared Error (RMSE)	$\$75,682,307.38$
Coefficient of Determination (R^2)	0.7271

5 Conclusion

The pipeline successfully extracted underlying structures within the movie metadata. K-Means and Hierarchical algorithms discovered distinct 3-cluster and 2-cluster segmentations, respectively.

For the predictive task, proper scaling of the target variable proved critical. The final neural network successfully explained 72.71% of the variance in movie revenue ($R^2 = 0.7271$). While the raw MSE appears massive (5.72×10^{15}), this is a mathematical consequence of squaring errors in the hundreds of millions. The root mean squared error (RMSE) translates to approximately \$75.7 million, which represents a highly effective and optimized model given the notorious volatility of box office revenues and the limited feature set available.